Variable Selection for Wasserstein Robust Hypothesis Testing

1 Problem formulation

Throughout this paper, we consider the variable selection problem for robust hypothesis testing. Let $[D] = \{1, \ldots, D\}$ and d be a positive integer denoting the sparsity budget. Given a detector $f : \mathbb{R}^D \to \mathbb{R}$, we define the worst-case risk as

$$R(f) = \sup_{\mathbb{P}_0 \in \mathcal{P}_0, \mathbb{P}_1 \in \mathcal{P}_1} \left\{ \mathbb{E}_{\mathbb{P}_0} \ell(-f(X)) + \mathbb{E}_{\mathbb{P}_1} \ell(f(X)) \right\}, \tag{1}$$

where $\mathcal{P}_k, k = 0, 1$ denote the Wasserstein distance-based ambiguity sets, and $\ell(\cdot)$ is the generating function that is nonnegative-valued, non-decreasing, convex, and satisfies $\ell(0) = 1, \lim_{t \to -\infty} \ell(t) = 0$. For $k = 0, 1$, assume we have n_k available observations in the group k, which are denoted as $\{x_i^{(k)}\}_{i=1}^{n_k}$. We denote the corresponding empirical distribution as $\widehat{\mathbb{P}}_k = \frac{1}{n_k} \sum_{i=1}^{n_k} \delta_{x_i^{(k)}}$, and construct the Wasserstein ambiguity set as

$$\mathcal{P}_k = \left\{ \mathbb{P}_k : \mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) \leq \rho_k \right\}.$$

Here, $\mathcal{W}_1(\cdot, \cdot)$ refers to the 1-Wasserstein distance equipped with the ℓ_1-norm.

For $S \subseteq [D]$, define the collection of detectors whose input only depends on coordinates belonging to S:

$$\mathcal{F}_S = \left\{ f : \mathbb{R}^D \to \mathbb{R}, f(x) = g(x[S]) \text{ for some measurable function } g : \mathbb{R}^{|S|} \to \mathbb{R} \right\}.$$

Note that for a given vector $x \in \mathbb{R}^D$, we use $x[i]$ to denote its i-th coordinate, and $x[S]$ to denote the concatenation of its coordinates whose indices belong to S. Our proposed variable selection problem can be formulated as the following combinatorial optimization problem

$$\min_{S \subseteq [D], |S| \leq d} \min_{f \in \mathcal{F}_S} R(f). \tag{2}$$

2 Convex formulation

For $S \subseteq [D]$, denote $V(S) = \min_{f \in \mathcal{F}_S} R(f)$ and define

$$\mathcal{X} = \{x_i^{(0)} : i \in [n_0]\} \cup \{x_j^{(1)} : j \in [n_1]\}.$$

Assume $n_0, n_1 \geq 1$, $\rho_0, \rho_1 > 0$, and that ℓ is finite on $\mathbb{R}$. The points of $\mathcal{X}$ serve as transport targets. For $y \in \mathcal{X}$, let $g_y \in \mathbb{R}$ denote the detector score at y, and write $g = (g_y)_{y \in \mathcal{X}}$. For masses $a, b \geq 0$, define

$$\psi(a, b) = \inf_{t \in \mathbb{R}} \left\{ a\ell(-t) + b\ell(t) \right\}.$$

This is the smallest loss at a point that carries mass a of class 0 and mass b of class 1. The function $p \mapsto \psi(1 - p, p)$ on $[0, 1]$ is the auxiliary function of [2], and ψ is positively homogeneous.

The next lemma is [2, Proposition 1 and Theorem 3] applied to the projected observations $x_i^{(k)}[S]$. The reduction to projections rests on two facts: projection does not increase ℓ_1 transport costs, and every transport between projected points lifts to a transport in $\mathbb{R}^D$ of the same cost that keeps the unselected coordinates of each observation.

Lemma 2.1 (Least favorable distributions)**.**

$$\begin{aligned} V(S) = \max_{p,\gamma\geq 0} \quad & \sum_{y\in\mathcal{X}} \psi\big(p_0(y), p_1(y)\big) \\ \textit{subject to} \quad & \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} = \frac{1}{n_k}, && i\in[n_k],\ k=0,1, \\ & p_k(y) = \sum_{i=1}^{n_k} \gamma_{i,y}^{(k)}, && y\in\mathcal{X},\ k=0,1, \\ & \sum_{i=1}^{n_k}\sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \leq \rho_k, && k=0,1. \end{aligned} \tag{3}$$

Here $\gamma_{i,y}^{(k)}$ is the mass transported from $x_i^{(k)}[S]$ to $y[S]$, and $p_k(y)$ is the mass of class k at the target y. Let Γ_S denote the set of transports $\gamma = (\gamma^{(0)}, \gamma^{(1)})$ that are feasible in Eq. (3) after p is eliminated.

Proposition 2.2 (Finite convex program)**.** *The value $V(S)$ equals the optimal value of the convex program*

$$\begin{aligned} \min_{g,a,\lambda} \quad & \frac{1}{n_0}\sum_i a_i^{(0)} + \frac{1}{n_1}\sum_j a_j^{(1)} + \rho_0\lambda_0 + \rho_1\lambda_1 \\ \textit{subject to} \quad & \ell(-g_y) \leq a_i^{(0)} + \lambda_0\|x_i^{(0)}[S] - y[S]\|_1, && i\in[n_0],\ y\in\mathcal{X}, \\ & \ell(g_y) \leq a_j^{(1)} + \lambda_1\|x_j^{(1)}[S] - y[S]\|_1, && j\in[n_1],\ y\in\mathcal{X}, \\ & g\in\mathbb{R}^{\mathcal{X}}, \quad a^{(k)}\in\mathbb{R}^{n_k}, \quad \lambda_k\geq 0, && k=0,1, \end{aligned} \tag{4}$$

and the minimum is attained.

In Eq. (4), λ_k is the price of the transport budget ρ_k. At an optimal solution, each $a_i^{(k)}$ takes its smallest feasible value

$$a_i^{(k)} = \max_{y\in\mathcal{X}}\big\{\ell\big((-1)^{k+1}g_y\big) - \lambda_k\|x_i^{(k)}[S] - y[S]\|_1\big\},$$

the largest loss that the observation $x_i^{(k)}$ can reach at a target, net of the transport price. Appendix A derives the program from Lemma 2.1 by Sion's minimax theorem [3] and linear programming duality.

A solution of Eq. (4) assigns scores only to the targets. To obtain a detector on $\mathbb{R}^{|S|}$, let (g, a, λ) be optimal and define, for $\xi\in\mathbb{R}^{|S|}$,

$$E_k(\xi) = \min_{i\in[n_k]}\big\{a_i^{(k)} + \lambda_k\|x_i^{(k)}[S] - \xi\|_1\big\}, \qquad k=0,1.$$

The constraints of Eq. (4) state that $\ell(-g_y) \leq E_0(y[S])$ and $\ell(g_y) \leq E_1(y[S])$ for every $y\in\mathcal{X}$. We extend the scores so that these bounds hold on all of $\mathbb{R}^{|S|}$:

$$\ell(-g(\xi)) \leq E_0(\xi), \qquad \ell(g(\xi)) \leq E_1(\xi), \qquad \xi\in\mathbb{R}^{|S|}. \tag{5}$$

For a general ℓ, Eq. (5) plays the role of the interval of optimal randomized tests in [4, Theorem 1].

We construct one such extension. If $y, y'\in\mathcal{X}$ and $y[S] = y'[S]$, the constraints of Eq. (4) on g_y and $g_{y'}$ coincide, so replacing $g_{y'}$ by g_y preserves feasibility and the objective. We therefore take $g_y = g_{y'}$ and set $g(y[S]) = g_y$. For $\xi\notin\{y[S] : y\in\mathcal{X}\}$, let i and j be the smallest indices attaining $E_0(\xi)$ and $E_1(\xi)$, respectively, and set

$$r_0 = \|x_i^{(0)}[S] - \xi\|_1, \qquad r_1 = \|x_j^{(1)}[S] - \xi\|_1,$$

$$\tau = \frac{r_0}{r_0 + r_1}, \qquad g(\xi) = (1-\tau)g_{x_i^{(0)}} + \tau g_{x_j^{(1)}}.$$

Both distances are positive because ξ is not a projected observation. Thus $g(\xi)$ is a convex combination of the scores at $x_i^{(0)}$ and $x_j^{(1)}$, with the larger weight on the point closer to ξ.

Proposition 2.3 (Optimal detector)**.** *The extended score g is Borel measurable and satisfies Eq. (5). The detector $f(x) = g(x[S])$ belongs to $\mathcal{F}_S$ and attains $R(f) = V(S)$.*

The proof in Appendix B uses the interpolation rule only to verify Eq. (5); every Borel extension that satisfies Eq. (5) is optimal.

To optimize the coordinate selection, define $z = \mathbf{1}_S$ and $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. For $\gamma = (\gamma^{(0)}, \gamma^{(1)})$ with $\gamma^{(k)} \in \mathbb{R}_+^{n_k \times |\mathcal{X}|}$, let $p_k(y) = \sum_i \gamma_{i,y}^{(k)}$ as in Eq. (3), and denote the objective of Eq. (3) by

$$H(\gamma) = \sum_{y \in \mathcal{X}} \psi\big(p_0(y), p_1(y)\big).$$

Since ψ is an infimum of linear functions with nonnegative coefficients, H is concave, nondecreasing and positively homogeneous: $H(s\gamma) = sH(\gamma)$ for $s \ge 0$. Define the average class mass and the class-k transport cost in coordinate m by

$$M(\gamma) = \frac{1}{2} \sum_{k=0}^{1} \sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)},$$

$$c_k(\gamma)[m] = \sum_{i=1}^{n_k} \sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} |x_i^{(k)}[m] - y[m]|, \qquad k = 0, 1,\ m \in [D].$$

For a scalar $t \in \mathbb{R}$, write $t_+ = \max\{t, 0\}$. The next result expresses the reciprocal of the selection objective as a minimum of functions affine in z.

Proposition 2.4 (Convex selection function)**.** *Let $\mathcal{D}$ be the set of pairs (γ, μ) with $\gamma \ge 0$ and $\mu \in \mathbb{R}_+^D$ such that*

$$\sum_{y \in \mathcal{X}} \gamma_{i,y}^{(k)} = \frac{M(\gamma)}{n_k}, \quad i \in [n_k], \qquad \sum_{m=1}^{D} \big(c_k(\gamma)[m] - \rho_k \mu_m\big)_+ \le \rho_k M(\gamma), \qquad k = 0, 1.$$

For $z \in [0,1]^D$, define $F(z) > 0$ by

$$\frac{1}{F(z)} = \min\big\{M(\gamma) + \mu^\top z :\ (\gamma, \mu) \in \mathcal{D},\ H(\gamma) \ge 1\big\}. \tag{6}$$

Then the minimum is attained, F is convex and nonincreasing with $F \le 2$, and

$$F(\mathbf{1}_S) = V(S), \qquad \min_{|S| \le d} V(S) = \min_{z \in \mathcal{Z}_d} F(z). \tag{7}$$

The row constraints of $\mathcal{D}$ require both classes to carry the same total mass $M(\gamma)$, which may differ from one; in Eq. (3), each class carries unit mass. Since $\mathcal{D}$ is a cone and H and M are positively homogeneous, Eq. (6) is equivalent to

$$F(z) = \max\left\{\frac{H(\gamma)}{M(\gamma) + \mu^\top z} :\ (\gamma, \mu) \in \mathcal{D},\ H(\gamma) > 0\right\}.$$

The variable μ_m is a price on coordinate m. Paying $\mu_m z_m$ in the objective exempts the first $\rho_k \mu_m$ units of the class-k cost in coordinate m from the budget constraint. At $z = \mathbf{1}_S$, the coordinates outside S are exempt at no charge, and only the cost on S remains constrained, as in Eq. (3).

Appendix C proves the proposition from Lemma 2.1. Minimizing F over $\mathcal{Z}_d$ therefore solves the original problem, and minimizing it over the continuous relaxation $\{z \in [0,1]^D : \mathbf{1}^\top z \le d\}$ gives a lower bound.

3 An outer-approximation approach

We apply outer approximation [1] to the concave function $1/F$. For fixed z, Eq. (6) minimizes a linear function over the convex set $\mathcal{D} \cap \{H \ge 1\}$. Let (γ^t, μ^t) be a minimizer at z^t, and define

$$c_t(z) = M(\gamma^t) + (\mu^t)^\top z. \tag{8}$$

The feasible set of Eq. (6) does not depend on z, so

$$\frac{1}{F(z)} \le c_t(z), \quad z \in [0,1]^D, \qquad \frac{1}{F(z^t)} = c_t(z^t).$$

Equivalently, $F(z) \ge 1/c_t(z)$. Because $t \mapsto 1/t$ is convex, $1/c_t$ dominates its linearization at z^t, which is the standard outer-approximation cut for F.

At a binary point $z^t = \mathbf{1}_{S^t}$, the cut requires no additional solve. Let $\gamma^\star$ be a least favorable transport of Eq. (3) at S^t. Appendix C.1 shows that the pair

$$\gamma^t = \frac{\gamma^\star}{V(S^t)}, \qquad \mu^t_m = \begin{cases} 0, & m \in S^t, \\ \max\limits_{k=0,1} \dfrac{c_k(\gamma^\star)[m]}{\rho_k V(S^t)}, & m \notin S^t, \end{cases}$$

is a minimizer of Eq. (6) at z^t.

After t evaluations, the next selection maximizes the smallest of the accumulated upper bounds on $1/F$:

$$\begin{aligned} (z^{t+1}, \theta_t) \in \operatorname*{arg\,max}_{z \in \mathcal{Z}_d,\, \theta} \quad & \theta \\ \text{subject to} \quad & \theta \le c_s(z), \qquad s = 1, \dots, t. \end{aligned} \tag{9}$$

The value $L_t = 1/\theta_t$ is a lower bound on $\mathrm{OPT} = \min_{z \in \mathcal{Z}_d} F(z)$, and the best evaluated value $U_t = \min_{s \le t} F(z^s)$ is an upper bound.

Algorithm 1 alternates between evaluating a selection and solving Eq. (9). It retains the best evaluated detector and stops when $U_t - L_t \le \varepsilon$. With exact solves, the algorithm terminates after finitely many iterations. Indeed, if $z^{t+1} = z^s$ for some $s \le t$, then

$$\theta_t \le c_s(z^{t+1}) = \frac{1}{F(z^s)} \le \frac{1}{U_t},$$

so $L_t \ge U_t$ and the stopping condition holds. Each nonterminal iteration therefore produces a new selection, and at most $|\mathcal{Z}_d|$ selections are evaluated. Since F is nonincreasing, the budget can also be imposed as $\mathbf{1}^\top z = d$ when $d \le D$.

Algorithm 1 Outer approximation for variable selection

Require: Samples $x_i^{(k)}$, radii $\rho_k > 0$, budget d, tolerance $\varepsilon > 0$, initial $z^1 \in \mathcal{Z}_d$

$U \leftarrow +\infty$, $t \leftarrow 1$
loop
  $S^t \leftarrow \{m : z^t_m = 1\}$
  Solve Eq. (4) at S^t for (g^t, a^t, λ^t), value $v_t = V(S^t)$, and a least favorable transport of Eq. (3)
  Form (γ^t, μ^t) from it and c_t by Eq. (8)
  if $v_t < U$ **then**
    $U \leftarrow v_t$; store $S^\star \leftarrow S^t$ and $(g^\star, a^\star, \lambda^\star) \leftarrow (g^t, a^t, \lambda^t)$
  end if
  Solve Eq. (9) with $c_1, \dots, c_t$ for (z^{t+1}, θ_t); $L_t \leftarrow 1/\theta_t$
  if $U - L_t \le \varepsilon$ **then**
    Extend $g^\star$ to $\mathbb{R}^{|S^\star|}$ as in Proposition 2.3
    return $S^\star$ and $f^\star(x) = g^\star(x[S^\star])$
  end if
  $t \leftarrow t + 1$
end loop

When a solver stops early, the bounds remain valid. Every feasible point of Eq. (4) gives an upper bound on $V(S^t)$, and every $(\gamma, \mu) \in \mathcal{D}$ with $H(\gamma) > 0$ gives the cut

$$\frac{1}{F(z)} \le \frac{M(\gamma) + \mu^\top z}{H(\gamma)}, \qquad z \in [0,1]^D.$$

For Eq. (9), take θ_t to be the solver's global upper bound.

A Proof of Proposition 2.2

Proof. For $g \in \mathbb{R}^{\mathcal{X}}$ and $\gamma \in \Gamma_S$, let

$$\Phi(g,\gamma) = \sum_{k=0}^{1} \sum_{i=1}^{n_k} \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)}\, \ell\big((-1)^{k+1} g_y\big) = \sum_{y\in\mathcal{X}} \big\{p_0(y)\ell(-g_y) + p_1(y)\ell(g_y)\big\}.$$

Exchange of maximum and infimum. The infimum of $\Phi(\cdot,\gamma)$ separates over the targets, and the infimum of the term at y over g_y is $\psi(p_0(y), p_1(y))$. Hence $\inf_g \Phi(g,\gamma)$ is the objective of Eq. (3) at γ, and Lemma 2.1 gives

$$V(S) = \max_{\gamma\in\Gamma_S} \inf_g \Phi(g,\gamma).$$

The set Γ_S is nonempty, convex and compact: it contains the transport that leaves every observation in place, it is defined by linear constraints, and its entries lie in $[0, \max_k 1/n_k]$. The function $\Phi(\cdot,\gamma)$ is convex and continuous, because ℓ is convex and finite and $\gamma \geq 0$; the function $\Phi(g,\cdot)$ is linear, hence concave and continuous. Sion's minimax theorem [3] therefore gives

$$V(S) = \inf_g \max_{\gamma\in\Gamma_S} \Phi(g,\gamma). \tag{10}$$

Duality for fixed scores. Fix g. The constraints of Γ_S do not couple the two classes, so the inner maximum in Eq. (10) is the sum over $k = 0, 1$ of the linear programs

$$\begin{aligned} \max_{\gamma^{(k)}\geq 0} \quad & \sum_{i=1}^{n_k} \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} \ell\big((-1)^{k+1} g_y\big) \\ \text{subject to} \quad & \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} = \frac{1}{n_k}, \qquad i \in [n_k], \\ & \sum_{i=1}^{n_k} \sum_{y\in\mathcal{X}} \gamma_{i,y}^{(k)} \|x_i^{(k)}[S] - y[S]\|_1 \leq \rho_k. \end{aligned}$$

With a free multiplier $a_i^{(k)}$ for the ith row constraint and a multiplier $\lambda_k \geq 0$ for the budget constraint, the dual linear program is

$$\begin{aligned} \min_{a^{(k)},\,\lambda_k\geq 0} \quad & \frac{1}{n_k} \sum_{i=1}^{n_k} a_i^{(k)} + \rho_k \lambda_k \\ \text{subject to} \quad & \ell\big((-1)^{k+1} g_y\big) \leq a_i^{(k)} + \lambda_k \|x_i^{(k)}[S] - y[S]\|_1, \qquad i \in [n_k],\ y \in \mathcal{X}. \end{aligned}$$

The primal program is feasible, since leaving every observation in place has cost $0 \leq \rho_k$, and its feasible set is bounded. By linear programming duality, the primal and dual optimal values are equal. Substituting the two dual programs into Eq. (10) and combining the infimum over g with the minima over (a,λ) shows that $V(S)$ is the infimum of Eq. (4).

Attainment. The point $g = 0$, $a_i^{(k)} = 1$, $\lambda = 0$ is feasible because $\ell(0) = 1$, and its objective value is 2. Let $\mathcal{K}$ be the set of feasible points with objective at most 2. Since the objective is continuous, it suffices to show that $\mathcal{K}$ is compact. The set $\mathcal{K}$ is closed because ℓ is continuous. Let $(g,a,\lambda) \in \mathcal{K}$. The constraint with source $x_i^{(k)}$ and target $y = x_i^{(k)}$ reads $\ell((-1)^{k+1} g_{x_i^{(k)}}) \leq a_i^{(k)}$, so $a^{(k)} \geq 0$. Every term of the objective is then nonnegative, and therefore

$$0 \leq a_i^{(k)} \leq 2n_k, \qquad 0 \leq \lambda_k \leq \frac{2}{\rho_k}.$$

With $\Delta = \max\{\|x[S] - y[S]\|_1 : x, y \in \mathcal{X}\}$, the constraints give, for every $y \in \mathcal{X}$,

$$\ell(-g_y) \leq 2n_0 + \frac{2\Delta}{\rho_0}, \qquad \ell(g_y) \leq 2n_1 + \frac{2\Delta}{\rho_1}.$$

Moreover, $\ell(t) \to \infty$ as $t \to \infty$. Indeed, since $\lim_{t\to-\infty} \ell(t) = 0 < 1$, there is $t_0 < 0$ with $\ell(t_0) < 1$, and convexity of ℓ applied to $t_0 < 0 < t$ gives

$$\ell(t) \geq \ell(0) + \frac{1 - \ell(t_0)}{-t_0}\, t, \qquad t > 0.$$

Hence g_y and $-g_y$ are bounded above on $\mathcal{K}$. Thus $\mathcal{K}$ is bounded, and therefore compact. □

B Proof of Proposition 2.3

Proof. Let $P = \{y[S] : y \in \mathcal{X}\}$ be the set of projected targets.

The bounds in Eq. (5) on P. Let $\xi = y[S]$ with $y \in \mathcal{X}$. The constraints of Eq. (4) with target y give, for every $i \in [n_0]$,

$$\ell(-g(\xi)) = \ell(-g_y) \leq a_i^{(0)} + \lambda_0 \|x_i^{(0)}[S] - \xi\|_1.$$

Minimizing over i gives $\ell(-g(\xi)) \leq E_0(\xi)$. The same argument for class 1 gives $\ell(g(\xi)) \leq E_1(\xi)$.

The bounds in Eq. (5) off P. Let $\xi \notin P$, let i, j, r_0, r_1 and τ be as in Section 2, and let

$$b = \|x_i^{(0)}[S] - x_j^{(1)}[S]\|_1.$$

The constraints of Eq. (4) with source $x_i^{(0)}$ and targets $x_i^{(0)}$, $x_j^{(1)}$, and with source $x_j^{(1)}$ and targets $x_j^{(1)}$, $x_i^{(0)}$, give

$$\begin{aligned} \ell(-g_{x_i^{(0)}}) &\leq a_i^{(0)}, & \ell(-g_{x_j^{(1)}}) &\leq a_i^{(0)} + \lambda_0 b, \\ \ell(g_{x_j^{(1)}}) &\leq a_j^{(1)}, & \ell(g_{x_i^{(0)}}) &\leq a_j^{(1)} + \lambda_1 b. \end{aligned} \tag{11}$$

The triangle inequality gives $b \leq r_0 + r_1$, hence

$$\tau b = \frac{r_0 b}{r_0 + r_1} \leq r_0, \qquad (1-\tau) b = \frac{r_1 b}{r_0 + r_1} \leq r_1.$$

By convexity of ℓ, Eq. (11), $\lambda_k \geq 0$, and the choice of i and j,

$$\ell(-g(\xi)) \leq (1-\tau)\ell(-g_{x_i^{(0)}}) + \tau\ell(-g_{x_j^{(1)}}) \leq a_i^{(0)} + \lambda_0 \tau b \leq a_i^{(0)} + \lambda_0 r_0 = E_0(\xi),$$

$$\ell(g(\xi)) \leq (1-\tau)\ell(g_{x_i^{(0)}}) + \tau\ell(g_{x_j^{(1)}}) \leq a_j^{(1)} + \lambda_1 (1-\tau) b \leq a_j^{(1)} + \lambda_1 r_1 = E_1(\xi).$$

Measurability. For $i \in [n_0]$ and $j \in [n_1]$, let A_{ij} be the set of $\xi \notin P$ for which i and j are the smallest indices attaining $E_0(\xi)$ and $E_1(\xi)$. Each A_{ij} is Borel, being defined by finitely many equalities and inequalities between continuous functions of ξ, and g is continuous on A_{ij}. The sets A_{ij} and the finitely many points of P partition $\mathbb{R}^{|S|}$, so g is Borel. Hence $f \in \mathcal{F}_S$.

The risk of f. Fix k and $\mathbb{P}_k \in \mathcal{P}_k$. By Eq. (5),

$$\ell\big((-1)^{k+1} f(x)\big) \leq E_k(x[S]), \qquad x \in \mathbb{R}^D.$$

Since $\|u[S] - v[S]\|_1 \leq \|u - v\|_1$ for $u, v \in \mathbb{R}^D$, each function

$$x \mapsto a_i^{(k)} + \lambda_k \|x_i^{(k)}[S] - x[S]\|_1, \qquad i \in [n_k],$$

is λ_k-Lipschitz for $\|\cdot\|_1$ on $\mathbb{R}^D$, and so is their minimum $x \mapsto E_k(x[S])$. This function is integrable under $\mathbb{P}_k$, which has a finite first moment because $\mathcal{W}_1(\widehat{\mathbb{P}}_k, \mathbb{P}_k) < \infty$. For every coupling π of $\widehat{\mathbb{P}}_k$ and $\mathbb{P}_k$,

$$\mathbb{E}_{\mathbb{P}_k} E_k(X[S]) = \int E_k(x'[S])\, d\pi(x, x') \leq \frac{1}{n_k} \sum_{i=1}^{n_k} E_k\big(x_i^{(k)}[S]\big) + \lambda_k \int \|x - x'\|_1\, d\pi(x, x').$$

Taking the infimum over couplings and using $E_k(x_i^{(k)}[S]) \leq a_i^{(k)}$, the ith term of the minimum that defines E_k, gives

$$\mathbb{E}_{\mathbb{P}_k} \ell\big((-1)^{k+1} f(X)\big) \leq \frac{1}{n_k} \sum_{i=1}^{n_k} a_i^{(k)} + \lambda_k \rho_k.$$

Summing over k and taking the supremum over $\mathcal{P}_0 \times \mathcal{P}_1$ bounds $R(f)$ by the objective of Eq. (4) at (g, a, λ), which equals $V(S)$ by Proposition 2.2. Conversely, $R(f) \geq V(S)$ because $f \in \mathcal{F}_S$. □

C Proof of Proposition 2.4

Three facts are used throughout. First, for $a, b \geq 0$,

$$2\min\{a, b\} \leq \psi(a, b) \leq a + b. \tag{12}$$

The upper bound takes $t = 0$ in the definition of ψ. For the lower bound, convexity gives $\ell(t) + \ell(-t) \geq 2\ell(0) = 2$, and since $\ell \geq 0$,

$$a\ell(-t) + b\ell(t) \geq \min\{a, b\}\big(\ell(-t) + \ell(t)\big) \geq 2\min\{a, b\}, \qquad t \in \mathbb{R}.$$

Summing the upper bound over the targets gives $H(\gamma) \leq 2M(\gamma)$. Second, H is upper semicontinuous, because ψ is an infimum of continuous functions. Third, the transport that leaves every observation in place,

$$\iota_{i,y}^{(k)} = \frac{1}{n_k}\mathbf{1}\{y = x_i^{(k)}\},$$

satisfies $M(\iota) = 1$ and $c_k(\iota) = 0$.

C.1 Equality at binary selections

Proof of $F(\mathbf{1}_S) = V(S)$. Fix S.

Positivity of $V(S)$. Let $b_0 = \|x_1^{(0)}[S] - x_1^{(1)}[S]\|_1$. Set $\varepsilon = 1/n_0$ if $b_0 = 0$, and $\varepsilon = \min\{1/n_0, \rho_0/b_0\}$ otherwise. Move mass ε of $x_1^{(0)}$ to the target $x_1^{(1)}$ and leave all other mass in place. This transport is feasible in Eq. (3), because its cost is $\varepsilon b_0 \leq \rho_0$, and it places class-0 mass at least ε and class-1 mass at least $1/n_1$ at $x_1^{(1)}$. Since ψ is nonnegative and nondecreasing, Eq. (12) gives

$$V(S) \geq \psi(\varepsilon, 1/n_1) \geq 2\min\{\varepsilon, 1/n_1\} > 0.$$

The bound $1/F(\mathbf{1}_S) \leq 1/V(S)$. The maximum in Eq. (3) is attained, because Γ_S is compact and H is upper semicontinuous. Let $\gamma^\star$ attain it. Since $\|x[S] - y[S]\|_1 = \sum_{m\in S} |x[m] - y[m]|$, the constraints of Eq. (3) give

$$M(\gamma^\star) = 1, \qquad H(\gamma^\star) = V(S), \qquad \sum_{m\in S} c_k(\gamma^\star)[m] \leq \rho_k, \quad k = 0, 1.$$

Define $\mu \in \mathbb{R}_+^D$ by

$$\mu_m = 0 \quad (m \in S), \qquad \mu_m = \max_{k=0,1} \frac{c_k(\gamma^\star)[m]}{\rho_k} \quad (m \notin S). \tag{13}$$

For $m \notin S$, this choice gives $(c_k(\gamma^\star)[m] - \rho_k\mu_m)_+ = 0$. Hence

$$\sum_{m=1}^{D} \big(c_k(\gamma^\star)[m] - \rho_k\mu_m\big)_+ = \sum_{m\in S} c_k(\gamma^\star)[m] \leq \rho_k = \rho_k M(\gamma^\star).$$

The row constraints of $\mathcal{D}$ hold because $M(\gamma^\star) = 1$, so $(\gamma^\star, \mu) \in \mathcal{D}$. Since $\mathcal{D}$ is a cone and H and M are positively homogeneous, $(\gamma^\star, \mu)/V(S)$ is feasible in Eq. (6) with $H = 1$, and its objective at $z = \mathbf{1}_S$ is

$$\frac{M(\gamma^\star) + \mu^\top \mathbf{1}_S}{V(S)} = \frac{1}{V(S)}.$$

The bound $1/F(\mathbf{1}_S) \geq 1/V(S)$. Let $(\gamma, \mu) \in \mathcal{D}$ with $H(\gamma) \geq 1$, and let $s = M(\gamma) + \mu^\top \mathbf{1}_S$. Since $H(\gamma) \leq 2M(\gamma)$, we have $M(\gamma) \geq 1/2$, and $s \geq M(\gamma) > 0$ because $\mu \geq 0$. Define

$$\gamma' = \frac{\gamma}{s} + \Big(1 - \frac{M(\gamma)}{s}\Big)\iota.$$

The coefficient of ι is nonnegative, so $\gamma' \geq 0$, and the rows of γ' sum to

$$\frac{M(\gamma)}{sn_k} + \Big(1 - \frac{M(\gamma)}{s}\Big)\frac{1}{n_k} = \frac{1}{n_k}.$$

The cost c_k is linear and $c_k(\iota) = 0$, so $c_k(\gamma') = c_k(\gamma)/s$. For every m,

$$c_k(\gamma)[m] \leq \big(c_k(\gamma)[m] - \rho_k\mu_m\big)_+ + \rho_k\mu_m.$$

Summing over $m \in S$ and using the cost constraint of $\mathcal{D}$,

$$\sum_{m\in S} c_k(\gamma)[m] \le \sum_{m=1}^{D} \big(c_k(\gamma)[m] - \rho_k \mu_m\big)_+ + \rho_k \mu^\top \mathbf{1}_S \le \rho_k \big(M(\gamma) + \mu^\top \mathbf{1}_S\big) = \rho_k s.$$

Hence $\sum_{m\in S} c_k(\gamma')[m] \le \rho_k$, and $\gamma' \in \Gamma_S$. Since $\gamma' \ge \gamma/s$ entrywise and H is nondecreasing and positively homogeneous,

$$V(S) \ge H(\gamma') \ge H(\gamma/s) = \frac{H(\gamma)}{s} \ge \frac{1}{s}.$$

Thus every feasible point of Eq. (6) at $z = \mathbf{1}_S$ has objective $s \ge 1/V(S)$.

The two bounds give $F(\mathbf{1}_S) = V(S)$. Moreover, $(\gamma^\star, \mu)/V(S)$, with μ given by Eq. (13), attains the minimum in Eq. (6); this is the minimizer used in Section 3. □

C.2 Attainment and convexity

Proof of the remaining claims. Fix $z \in [0,1]^D$.

Attainment and the bound $F \le 2$. The feasible set of Eq. (6) is nonempty by Appendix C.1. Since $\mu, z \ge 0$ and $H(\gamma) \le 2M(\gamma)$, every feasible point satisfies

$$M(\gamma) + \mu^\top z \ge M(\gamma) \ge \frac{1}{2}.$$

Let (γ^j, μ^j) be a minimizing sequence with objective values at most $B < \infty$. Then $M(\gamma^j) \le B$, and each entry of γ^j is at most the total mass $M(\gamma^j)$ of its class, so the sequence γ^j is bounded. Replacing μ^j_m by

$$\min\Big\{\mu^j_m,\ \max_{k=0,1} \frac{c_k(\gamma^j)[m]}{\rho_k}\Big\}$$

changes μ^j_m only if $\rho_k \mu^j_m \ge c_k(\gamma^j)[m]$ for both k, and then the terms $(c_k(\gamma^j)[m] - \rho_k \mu^j_m)_+$ are zero before and after the replacement. The replacement therefore keeps the point in $\mathcal{D}$, and it does not increase the objective because $z \ge 0$. We may thus assume that μ^j is bounded. A subsequence converges to some (γ, μ). The limit lies in $\mathcal{D}$, which is closed because its constraints involve only continuous functions, and it satisfies $H(\gamma) \ge 1$ because H is upper semicontinuous. The objective is continuous, so the limit attains the minimum. Hence $1/F(z) \ge 1/2$, that is, $F(z) \le 2$.

Convexity and monotonicity. The set $\mathcal{D} \cap \{H \ge 1\}$ is convex. Indeed, the row constraints are linear equations; the cost constraint bounds a convex function of (γ, μ), a sum of positive parts of affine functions, by the linear function $\rho_k M(\gamma)$; and $\{H \ge 1\}$ is a superlevel set of a concave function. This set does not depend on z, and at each of its points the objective $M(\gamma) + \mu^\top z$ is affine in z with slope $\mu \ge 0$. Therefore $1/F$, a pointwise minimum of such functions, is concave, nondecreasing and positive on $[0,1]^D$. Since $t \mapsto 1/t$ is convex and decreasing on $(0, \infty)$, F is convex and nonincreasing.

Proof of Eq. (7). The first identity is proved in Appendix C.1. The second follows because $S \mapsto \mathbf{1}_S$ is a bijection from $\{S \subseteq [D] : |S| \le d\}$ onto $\mathcal{Z}_d$. □

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